

B.Tech. Degree I Semester Regular Examination November 2023

CE/ME/SE 23-200-0106 A ENVIRONMENTAL AND LIFE SCIENCES
(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Identify the global environmental issues.

CO2: Examine the types of pollution in society along with their sources.

CO3: Elucidate the basic biological concept via relevant industrial applications and case studies.

CO4: Evaluate the principles of design and development, for exploring novel bioengineering projects.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

	(5 × 2 = 10)	Marks	BL	CO	PO
I. (a)	Mention some consumptive uses of biodiversity.	2	L1	1	2
(b)	What role does biodiversity play in the economic development of mankind?	2	L2	3	2
(c)	Write a short note on the methods of land reclamation.	2	L2	2	2
(d)	What is the role of pacemaker and defibrillator?	2	L4	4	2
(e)	Explain in short, the functioning of lungs.	2	L3	4	2

PART B

(4 × 10 = 40)

II.	Discuss the structure and function of an ecosystem.	10	L1	1	2
OR					
III.	Explain the structure of a grassland ecosystem.	10	L1	1	2
IV.	Explain the various effluent treatment processes.	10	L2	2	2
OR					
V.	Write a short note on the management of:				
	(i) E-waste	5	L2	2	2
	(ii) Drought.	5	L2	2	2
VI.	Mention the application of:				
	(i) Glucose oxidase in biosensor	5	L3	3	2
	(ii) Lipids as cleansers.	5	L3	3	2
OR					
VII.	Write short notes on:				
	(i) Friction reducing shark skin	5	L3	3	2
	(ii) Aerodynamic design of birds.	5	L3	3	2
VIII.	Explain the working of				
	(i) EEG	5	L4	4	2
	(ii) ECG.	5	L4	4	2
OR					
IX.	Discuss the application of various bio printing techniques for 3D printing of ears and food.	10	L4	4	2

Bloom's Taxonomy Levels

L1 - 24%, L2 - 26%, L3 - 25%, L4 - 25%.

